

Aditya Singh

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SUMMARY

Backend & AI Systems Engineer (final-year CSE) with production experience in distributed systems, agentic AI, and real-time infrastructure. Built a sub-millisecond HFT matching engine in Go and a Claude-powered autonomous AI agent with a live MCP server — both shipped end-to-end. Proficient in Python, Go, FastAPI, PostgreSQL, Redis, Kafka, gRPC, Docker, and AWS. Seeking SDE or AI-adjacent roles to build intelligent, high-performance backend systems.

EDUCATION

Vellore Institute of Technology, Bhopal

B.Tech in Computer Science and Engineering | CGPA: 7.80

Bhopal, India

Oct. 2022 – Present

EXPERIENCE

Research Intern

Jul. 2025 – Sep. 2025

IIT Ropar

- Sole developer on a production full-stack platform (React + Tailwind CSS) shipped to 200+ users — owned API design, database schema, and deployment end-to-end; Python data pipeline processing 500+ records at 95%+ accuracy
- Diagnosed a recurring workflow failure, shipped a fix, and wrote a post-mortem adopted by the team; automated verification workflows reducing manual effort by ~70%

PROJECTS

QuantFlow – AI-Native HFT Platform | *Go, Python, FastAPI, Kafka, gRPC, Docker, MCP, Playwright* 2026

- Engineered a production-grade microservices HFT platform — Go matching engine with goroutines and FIFO order book exposed via gRPC + Protobuf achieving sub-millisecond latency at 500K+ orders/sec; FastAPI order service with JWT auth, Pydantic validation, pre-trade risk engine, PostgreSQL via SQLAlchemy ORM, Alembic migrations, and WebSocket connection manager streaming live order book (100ms gRPC polling) and instant trade confirmations
- Designed event-driven architecture on Apache Kafka across 3 topics (`orders.placed`, `trades.executed`, `orders.cancelled`) with symbol-keyed partitioning; portfolio microservice consumes trade events to maintain real-time positions, weighted average cost, and realised P&L per user; containerised all 7 services with Docker Compose including health checks and dependency ordering
- Instrumented FastAPI and Go services with Prometheus metrics (counters, histograms, gauges) and Grafana dashboards tracking orders/sec, p99 matching latency, error rates, and Kafka consumer lag; distributed tracing via Jaeger and OpenTelemetry; custom MCP server (Anthropic SDK) exposing `place_order`, `get_portfolio`, `analyze_risk` — Claude executes real trades via natural language; CI/CD via GitHub Actions with Playwright E2E covering full order lifecycle and WebSocket assertions

Gravitas – Autonomous AI Task Manager | *FastAPI, PostgreSQL, Redis, Claude API, MCP, OAuth 2.0* 2026

- Built a production full-stack application — async FastAPI backend, React frontend, PostgreSQL with Alembic migrations, Redis caching, and JWT auth with refresh token rotation; 15+ endpoints documented via OpenAPI/Swagger
- Integrated an autonomous Claude API agent with a self-directed tool-use loop — agent reads context, calls backend APIs, and generates prioritised work plans without user intervention; exposed all functionality as an MCP server with 7 tools enabling Claude Desktop and Claude Code to control the app via natural language
- Engineered GitHub OAuth 2.0 with HMAC-SHA256 verified webhook ingestion and a `BaseProvider` abstraction supporting GitHub, Slack, and Linear — zero core changes per new integration; 20+ pytest suite with >90% coverage on auth, CRUD, and security flows

TECHNICAL SKILLS

Languages & Frameworks: Python, Go, JavaScript, SQL, Bash | FastAPI, Flask, React, Tailwind CSS, SQLAlchemy, asynpg
Distributed Systems: Apache Kafka, gRPC, Protobuf, WebSockets, Event Sourcing, REST APIs, OAuth 2.0, Webhooks, RBAC, JWT

AI & Agents: Anthropic Claude API, MCP Server (FastMCP), Agentic Tool-Use Loops, Prompt Engineering, Claude Code

Databases & Caching: PostgreSQL, TimescaleDB, Redis (Sorted Sets, TTL, Pub/Sub), Elasticsearch, Alembic, Query Optimisation

DevOps & Testing: Docker, GitHub Actions CI/CD, Playwright, pytest, Prometheus, Grafana, AWS, Nginx, Kubernetes (K3s), Git